

ED ZANELLI

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PROFESSIONAL SUMMARY

Technology executive with 30+ years of experience leading software engineering, data architecture, and cloud platform development. Founder of SeniorMinder, an AI-powered healthcare platform leveraging conversational AI, multimodal interfaces, and cloud-native architecture to improve senior safety and independence. Experienced in building AI-enabled products from concept through working prototype while leading engineering organizations.

SELECTED TECHNOLOGIES

OpenAI, Whisper, HeyGen LiveAvatar, ElevenLabs, Claude Code, Node.js, TypeScript, Electron, PostgreSQL, AWS, Git, REST APIs

PROFESSIONAL EXPERIENCE

SeniorMinder (Huntington Beach, CA)

Founder | April 2025 – Present

- Founded and led development of an AI-powered, HIPAA-aware platform that helps seniors remain independent while providing caregivers with proactive wellness and safety insights.
- Defined the product vision, AI architecture, technical roadmap, and user experience from concept through working prototype.
- Architected and implemented the complete software platform, including cloud infrastructure, conversational AI services, and caregiver applications.
- Designed and implemented voice biomarker analysis that extracts and evaluates speech characteristics such as cadence, clarity, pitch, tone, volume, and inflection to identify potential changes in cognitive and physical well-being.
- Architected a multimodal AI framework integrating **OpenAI**, **Whisper**, **HeyGen LiveAvatar**, and **ElevenLabs** to deliver natural, voice-driven interactions with automatic fallback from avatar to voice-only conversations.
- Integrated OpenAI language models to interpret natural conversations, assess wellness status, identify meaningful events, and generate proactive caregiver notifications.
- Implemented speech recognition and transcription using OpenAI Whisper, enabling accurate analysis of natural spoken conversations.
- Built secure cloud services using **AWS**, **Node.js**, **PostgreSQL**, and REST APIs for managing HIPAA-sensitive healthcare information.
- Leveraged Claude Code as an AI pair-programming companion for code review, refactoring, documentation, and design exploration while personally architecting, implementing, debugging, testing, and maintaining the production codebase.

Datagusto (London, UK)

Advisor | July 2024 – June 2025

- Coached and mentored CEO and CTO.
- Reviewed and advised regarding data architecture and framework, technical documentation, prototypes, demonstrations and presentations.

Crux Informatics (San Francisco, CA)

Director of Data Engineering | March 2022 – April 2024

- Led and coached 18 domestic employees and 48 offshore consultants responsible for building and maintaining data pipelines, curating metadata, and transforming data.
- Expanded company portfolio to include 60,000 active data pipelines from vendors such as Bloomberg, Dow Jones, FactSet, Morningstar, Refinitiv, SoftBank, and Thomson Reuters.

- Led the data engineering relationship with key accounts including Capital Group, Goldman Sachs and Two Sigma.
- Oversaw the creation and monitoring of a data engineering capacity model that allows the team to shift resources for on-time delivery, improving service level agreement (SLA) compliance from 33% to 100%.
- Identified and mitigated the main source of pipeline disruptions by prototyping schema protection in tooling that allowed pipelines to continue running despite schema changes such as added columns, dropped columns, and data type changes.
- Developed and presented a risk mitigation strategy that involved using existing algorithms to establish the schema that is stored in metadata, then importing data as text fields, allowing non-compliant data to be preserved in raw form. This strategy allows for all data to be captured while also allowing teams to establish policies for data transformation.
- Established and optimized pipeline delivery process, ensuring uniform approach to the design, development, deployment, and dispatching of data.
- Assessed the feasibility, cost, and transition process of adopting and implementing new technologies.
- Established automated workflows and improved tooling to decrease pipeline issues and increase productivity for all data engineers.
- Built and maintained relationships with clients to identify and resolve data pipeline issues, providing data-driven solutions to both clients and internal leadership.
- Led strategic approach to troubleshooting that allowed less technical teams to resolve commonly occurring issues.
- Collaborated with client counterparts to ensure delivered data was in formats that aligned with their workflows, including presenting new functionality internally to mitigate similar issues with future clients.

Vital Data Technology (Newport Beach, CA)

Director of Data Services | October 2019 – March 2022

- Managed a team of 9 responsible for data integration, data architecture, data governance, and the development of HIPAA-compliant applications.
- Developed and executed on engineering roadmap and projects for clients such as Arkansas Blue Cross/Blue Shield.
- Developed and maintained high-trust, HIPAA-compliant software for health plan customers using the Microsoft Azure platform and tools.
- Optimized HEDIS calculations for CountyCare (400K members), cutting end-to-end processing time from 4+ days to less than 5 hours by creating a multi-threaded application to run HEDIS in parallel rather than in series and increasing workflow efficiencies by up to 33%.
- Collaborated closely with Project Manager to design Jira dashboards and widgets that showcased work and provided detailed views to leadership on progress.
- Developed a COVID dashboard to chart test results and outcomes by employer, location, and other factors that helped identify clusters of positive test results.
- Proposed a rewrite of HEDIS measures to bridge the gap between arriving at the correct answer and performing optimally, enabling the team to run HEDIS for larger organizations.
- Led team in establishing shared secure file transfer protocol for importing health plan data, as well as creating a validation program to ensure all data was accurate and complete.
- Collaborated closely with Product and Data Science teams to design and develop new features that were ultimately presented as opportunities for future roadmaps.

Airship (Portland, OR)

Data Architect | January 2017 – August 2019

- Led a team of 3 data engineers to design and develop a cloud-based (AWS/RDS) business systems data warehouse encompassing financial, marketing, and customer usage data.
- Directed integration of data from cloud-based internal source systems (Salesforce, NetSuite, Zuora and Pardot) using FiveTran, including using security features.
- Developed, maintained, and orchestrated extract, transform, and load (ETL) programs and BI reporting for Finance and Marketing.
- Integrated post-acquisition customer data from AccEngage into the Enterprise Data Warehouse.

- Collaborated with the Finance team to support business unit needs.

Citrix (Goleta, CA)

Data Architect | January 2011 – December 2016

- Designed and developed ETL programs from raw data (Salesforce, GoToMeeting and GoToMyPC usage) into the enterprise data warehouse.
- Supervised a remote IT and QA team in Bangalore, India; performed code reviews, authored software coding standards and mentored developers for onshore and offshore teams.
- Developed online analytical processing (OLAP) cube for marketing leads and created self-service PowerPivot dashboards for stakeholders.
- Collaborated with Marketing and Sales to track marketing spend on a campaign basis to associate leads and opportunities with the campaign that generated them and tracked existing customer usage to forecast customer retention and upsell opportunities.

FURTHER PROFESSIONAL EXPERIENCE

Oracle Corporation | Sr. Development Manager

Redwood Shores, CA

Siebel Systems | Sr. Software Engineer

San Mateo, CA

EDUCATION

Bachelor of Science, Engineering and Applied Science, Economics | California Institute of Technology - Pasadena, CA